Goal: Be able to undo the process of differentiation.

$\int kf(x)\,dx = kF(x) + C$	$\int (f(x) + g(x))\,dx = F(x) + G(x) + C$
$\int f(g(x))g'(x)\,dx = F(g(x)) + C$	$\int (f(x) - g(x))\,dx = F(x) - G(x) + C$
$\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$ (when $n \neq -1$)	$\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$
$\int e^x\,dx = e^x + C$	$\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C$
$\int \sin x\,dx = -\cos x + C$	$\int \cos x\,dx = \sin x + C$
$\int \sec^2 x\,dx = \tan x + C$	$\int \sec x \tan x\,dx = \sec x + C$
$\int \csc x \cot x\,dx = -\csc x + C$	$\int \csc^2 x\,dx = -\cot x + C$
$\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1} x + C$	$\int \frac{1}{1+x^2}\,dx = \tan^{-1} x + C$

Table 1: Table of Antiderivatives

$f' \leftarrow f \leftarrow F$

Antiderivatives

Definition. A function F is an antiderivative of f on an interval I if $F'(x) = f(x)$ for all x an I. (If I is not specified, then it is just the domain of f).

Example 1. Consider the function $f(x) = x$:

1. Is $F(x) = \frac{1}{2}x^2$ an antiderivative of x (on $I = (-\infty, \infty)$)?

$F'(x) = \left[\frac{1}{2}x^2\right]' = \frac{1}{2} \cdot 2x = x$ ✓

2. Is $F(x) = x^2$ an antidervative of x?

$F'(x) = [x^2]' = 2x \neq x$

$2x \neq x$ for all x in $(-\infty, \infty)$

$f(x) = x$

$F'(x) = f(x)$

3. Is $F(x) = \frac{1}{2}x^2 + 1$ an antiderivative of x?

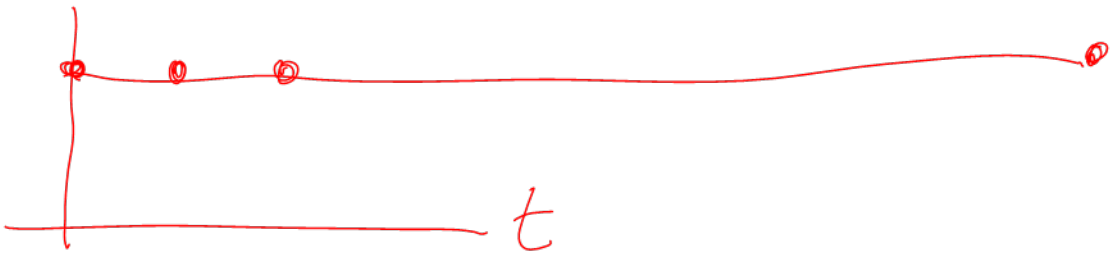
$F'(x) = \left[\frac{1}{2}x^2 + 1\right]' = \frac{1}{2} \cdot 2x + 0 = x$

Conclusion: Antiderivatives are not unique

$\frac{1}{2}x^2 + [\text{constant}]$

Theorem. If $F' = 0$ on an interval, then F is constant on the interval.

WHY?



Corollary. If F and G are antiderivatives of a function f, then

$$F(x) = G(x) + C$$

for some constant C.

WHY?

$$[F(x) - G(x)]' = F'(x) - G'(x) = f(x) - f(x) = 0$$

$$\Rightarrow F(x) - G(x) = [\text{constant}]$$

Indefinite Integration

Definition. The family of antiderivatives of a function f is called the (indefinite) integral of f, denoted

$$\int f(x)\,dx \overset{\text{BY THE COROLLARY}}{=\!=\!=} F(x) + C$$

Integral sign; Integrand; Integration Variable; Particular Antiderivative; Constant of Integration

Example 2. Find all possible antiderivatives of $f(x) = \sin x$.

Step 1: Find a particular antiderivative

$$[\cos x]' = -\sin x$$

$$(-1)[\cos x]' = \sin x$$

$$[-\cos x]' = \sin x$$

Step 2: $\int \sin x\,dx = -\cos x + C$

Example 3. Compute $\int \frac{1}{x}\,dx$ on $(0,\infty)$

Step 1: $[\ln x]' = \frac{1}{x}$

Step 2: $\int \frac{1}{x}\,dx = \ln x + C$

Theorem. In general, on $(-\infty,0)\cup(0,\infty) = \left\{\text{domain of } \frac{1}{x}\right\}$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

Example 4. Compute $\int x^2\,dx$

Step 1:

$$[x^3]' = 3x^2$$
$$\frac{1}{3}[x^3]' = x^2$$
$$\left[\tfrac{1}{3}x^3\right]' = x^2$$

Step 2: $\int x^2\,dx = \frac{1}{3}x^3 + C$

$[x^n]' = nx^{n-1}$

Theorem (Reverse Power Rule)**.** When $n \neq -1$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Example 5.

$$\int \sqrt{x}\,dx = \int x^{\frac{1}{2}}\,dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C$$
$$= \frac{2}{3}x^{\frac{3}{2}} + C$$

Theorem. Integrals are linear in the sense that:

(i) $\int (f(x) \pm g(x))\, dx = \int f(x)\, dx \pm \int g(x)\, dx$

(ii) If k is a constant, $\int kf(x)\, dx = k \int f(x)\, dx$

Example 6. Compute $\int \dfrac{2x^5 - \sqrt{x}}{x}\, dx$

$$\int \frac{2x^5 - \sqrt{x}}{x}\, dx = \int \left(\frac{2x^5}{x} - \frac{\sqrt{x}}{x^1}\right) dx \qquad \sqrt{x} = x^{\frac{1}{2}}$$

$$= \int (2x^4 - x^{-\frac{1}{2}})\, dx$$

$$= \int 2x^4\, dx - \int x^{-\frac{1}{2}}\, dx$$

$$= 2\int x^4\, dx - \int x^{-\frac{1}{2}}\, dx$$

$$= 2 \cdot \frac{x^{4+1}}{4+1} - \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C$$

$$= \frac{2}{5}x^5 - \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= \frac{2}{5}x^5 - 2\sqrt{x} + C$$

Check:

$$\frac{2x^5 - \sqrt{x}}{x}$$

$$2x^4 - \frac{1}{\sqrt{x}}$$

$$\frac{2x^4\sqrt{x} - 1}{\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}}$$

$$2x^4 - 2 \cdot \frac{1}{2\sqrt{x}}$$

Remark 1 (Notation): Add parentheses when the integrand has more than one term:

$$\int x^2 + x^3\, dx \longleftarrow \text{Not Quite Right}$$

$$\int (x^2 + x^3)\, dx \longleftarrow \text{Correct}$$

Remark 2: When dealing with multiple integrals, it is common practice to have only one constant of integration and to tack it on at the very end. We will demonstrate why in the next example.

Example 7. Compute $\int \sec x(\sec x + \cos x)\, dx$

$$\int \sec x(\sec x + \cos x)\,dx = \int (\sec^2 x + \sec x \cos x)\,dx$$

$$\sec x = \frac{1}{\cos x}$$

$$= \int (\sec^2 x + 1)\,dx = \int \sec^2 x\,dx + \int 1 x^0\,dx$$

$$\frac{x^{0+1}}{0+1} = x$$

$$= \tan x + C_1 + x + C_2$$

$$= \tan x + x + [C_1 + C_2]$$

$$= \tan x + x + C$$

Example 8. Compute $\int \frac{14 + 4xe^x}{x}\, dx$

$$\int \frac{14 + 4xe^x}{x}\,dx = \int \left(\frac{14}{x} + \frac{4xe^x}{x}\right) dx$$

$$= \int \left(\frac{14}{x} + 4e^x\right) dx = \int \frac{14}{x}\,dx + \int 4e^x\,dx$$

$$= 14 \int \frac{1}{x}\,dx + 4 \int e^x\,dx$$

$$= 14 \ln|x| + 4e^x + C$$

A Glimpse into MA 16020

We want techniques to "undo" the use of the Product, Quotient, and Chain Rule. **You won't need this for 16010**, but here is an example of such a thing:

Theorem (Reverse Chain Rule/"U-substitution")**.** If F is an antiderivative for f. Then,

$$\int f'(g(x))\, g'(x)\, dx = f(g(x)) + C$$

$$\int f(g(x)) \cdot g'(x) dx = F(g(x)) + C$$

$$f'(g(x)) \cdot g'(x) = \left[f(g(x))\right]'$$

Example 9. Compute $\int \cos(2x)\ dx$

$$\left[\sin(2x)\right]' = \underbrace{\cos(2x)}_{f'(g(x))} \cdot \underbrace{2}_{g'(x)}$$

$$\int \cos(2x)\, dx = \frac{1}{2}\int \cos(x) \cdot 2\, dx$$

$$= \frac{1}{2}\sin(2x) + C$$